

Spring 1980

Problem 1 (Sp80). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the unique function such that $f(x) = x$ if $-\pi \leq x < \pi$ and $f(x + 2n\pi) = f(x)$ for all $n \in \mathbb{Z}$.

1. Prove that the Fourier series of f is $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} 2 \sin nx}{n}$
2. Prove that the series does not converge uniformly.
3. For each $x \in \mathbb{R}$, find the sum of the series.

Problem 2 (Fa77, Sp80, Sp20). Let $f_n : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable for each $n = 1, 2, \dots$ with $|f'_n(x)| \leq 1$ for all n and x . Assume

$$\lim_{n \rightarrow \infty} f_n(x) = g(x)$$

for all x . Prove that $g : \mathbb{R} \rightarrow \mathbb{R}$ is continuous.

Problem 3 (Sp80). Let P_2 denote the set of real polynomials of degree ≤ 2 . Define the map $J : P_2 \rightarrow \mathbb{R}$ by

$$J(f) = \int_0^1 f(x)^2 dx$$

Let $Q = \{f \in P_2 \mid f(1) = 1\}$. Show that J attains a minimum value on Q and determine where the minimum occurs.

Problem 4 (Sp80, Fa22). Let $a > 0$ be a constant $\neq 2$. Let C_a denote the positively oriented circle of radius a centered at the origin. Evaluate

$$\int_{C_a} \frac{z^2 + e^z}{z^2(z-2)} dz$$

Problem 5 (Sp80). Let

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

be an analytic function in the open unit disc $\mathbb{D}$. Assume that

$$\sum_{n=2}^{\infty} n|a_n| \leq |a_1| \quad \text{with} \quad a_1 \neq 0$$

Prove that f is injective.

Problem 6 (Sp80). G is a group of order n , H a proper subgroup of order m , and $(n/m)! < 2n$. Prove G has a proper normal subgroup different from the identity.

Problem 7 (Sp80, Fa08). Let $n \geq 2$ be an integer such that $2^n + n^2$ is prime. Show that $n \equiv 3 \pmod{6}$.

Problem 8 (Sp80, Sp10, Fa13, Sp18). Let A and B be $n \times n$ complex matrices. Prove or disprove each of the following statements:

1. If A and B are diagonalizable, so is $A + B$.
2. If A and B are diagonalizable, so is AB .
3. If $A^2 = A$, then A is diagonalizable.
4. If A is invertible and A^2 is diagonalizable, then A is diagonalizable.
5. If AB is diagonalizable and invertible then so is BA .

Problem 9 (Sp80). Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Show that every real matrix B such that $AB = BA$ has the form $sI + tA$, where $s, t \in \mathbb{R}$.

Problem 10 (Sp80). Consider the differential equation $x' = \frac{x^3 - x}{1 + e^x}$.

1. Find all its constant solutions.
2. Discuss $\lim_{t \rightarrow \infty} x(t)$, where $x(t)$ is the solution such that $x(0) = \frac{1}{2}$.

Problem 11 (Sp80). Let $\mathcal{S} = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$ denote the unit sphere in $\mathbb{R}^3$. Evaluate the surface integral over $\mathcal{S}$:

$$\iint_{\mathcal{S}} (x^2 + y + z) \, dA$$

Problem 12 (Sp80). For any matrix $A \in M_{3 \times 3}$, define the linear operator $L_A : M_{3 \times 3} \rightarrow M_{3 \times 3}$, $L_A(B) = AB$. Suppose that the determinant of A is 32 and the minimal polynomial is $(t - 4)(t - 2)$. What is the trace of L_A ?

Problem 13 (Sp80). Let G be a subgroup of S_n . Assume G is *transitive*, that is, for any x and y in $\Omega = \{1, 2, \dots, n\}$, there is some $\sigma \in G$ with $\sigma(x) = y$.

1. Prove that n divides the order of G .
2. Suppose $n = 4$. For which integers $k \geq 1$ can such a G have order $4k$?

Problem 14 (Sp80). Find a real matrix B such that

$$B^4 = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & -1 & 1 \end{pmatrix}$$

Problem 15 (Sp80, Fa89). Show that a vector space over an infinite field cannot be the union of a finite number of proper subspaces.

Problem 16 (Sp80, Fa92). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be continuously differentiable. Assume the Jacobian matrix $(\partial f_i / \partial x_j)$ has rank n everywhere. Suppose f is *proper*, that is, $f^{-1}(K)$ is compact whenever K is compact. Prove $f(\mathbb{R}^n) = \mathbb{R}^n$.

Problem 17 (Sp80, Sp17). 1. Exhibit an element of S_9 of order 20.

2. Prove that no element of S_9 has order 18.

Problem 18 (Sp80). For each $t \in \mathbb{R}$, let $P(t)$ be a symmetric real $n \times n$ matrix whose entries are continuous functions of t . Suppose for all t that the eigenvalues of $P(t)$ are all ≤ -1 . Let $x(t) = (x_1(t), \dots, x_n(t))$ be a solution of the vector differential equation

$$\frac{dx}{dt} = P(t)x$$

Prove that

$$\lim_{t \rightarrow \infty} x(t) = 0$$

Problem 19 (Sp80). Let

$$f(z) = \sum_{n=0}^{\infty} c_n z^n$$

be analytic in the disc $\mathbb{D} = \{z \in \mathbb{C} \mid |z| < 1\}$. Assume f maps $\mathbb{D}$ one-to-one onto a domain G having area A . Prove

$$A = \pi \sum_{n=1}^{\infty} n |c_n|^2$$

Problem 20 (Sp80). Does there exist an analytic function mapping the annulus

$$A = \{z \mid 1 \leq |z| \leq 4\}$$

onto the annulus

$$B = \{z \mid 1 \leq |z| \leq 2\}$$

and taking $C_1 \rightarrow C_1, C_4 \rightarrow C_2$, where C_r is the circle of radius r ?